
KTU NOTES

MODULE 4

Tests on Induction motor, Circle diagram

Cogging & Crawling

Double cage Induction motor.

Starting of Induction motors – DOL starter, Autotransformer starter, Star-delta starter, Rotor resistance starter.

Braking of Induction motor – Plugging, Dynamic braking, Regenerative braking.

Speed control – Stator voltage control, V/F control, Rotor resistance control

CIRCLE DIAGRAM

- A **circle diagram** is a graphical representation of the performance of an electrical machine.
- It is commonly used to illustrate the performance of transformers, alternators, synchronous motors, and induction motors.
- It is very useful to study the performance of an electric machine under a large variety of operating conditions.
- The diagrammatic representation of a circle diagram makes it much easier to understand and remember compared to theoretical and mathematical descriptions.
- **No load test and blocked rotor test in an induction motor is required for drawing circle diagram.**

Construction of Circle diagram

- Data required for construction of circle diagram is obtained from No load test, Blocked rotor test & Stator resistance measurement test.
- Different steps involved in construction of circle diagram & calculations are given below.

<u>Step 1</u> - From No load test, calculate No load current/phase (I_{0ph}) & No load power factor angle (ϕ_0)

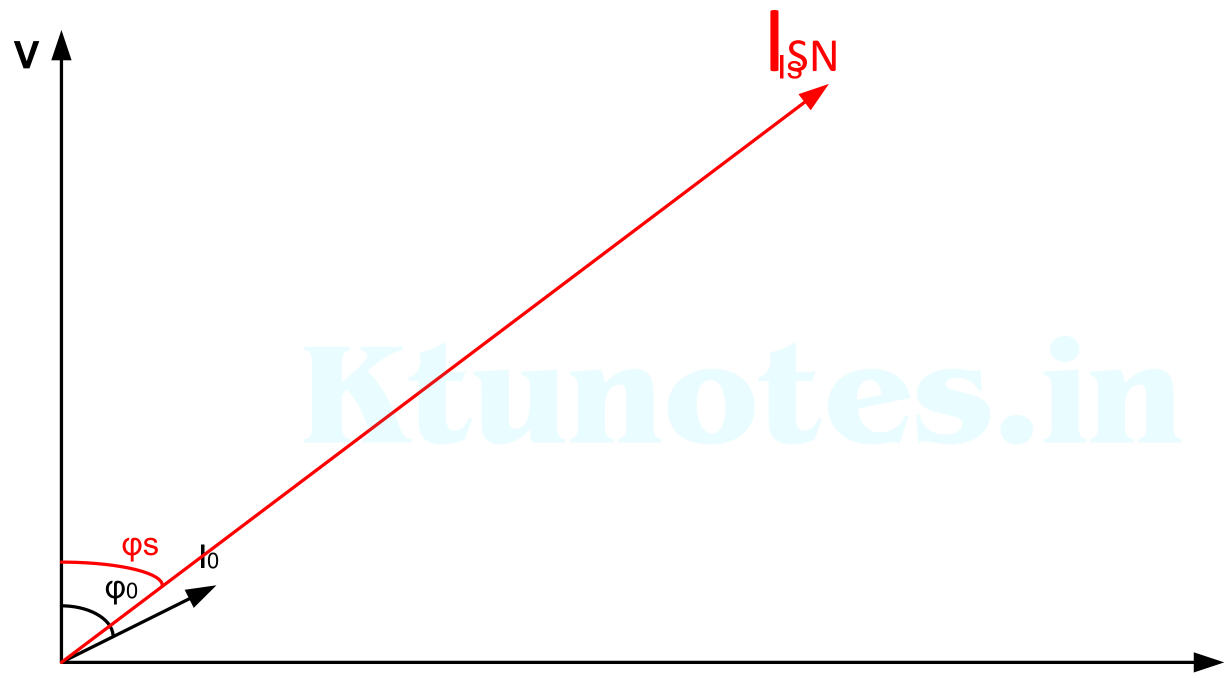
$$\phi_0 = Cos^{-1}\left(\frac{P_0}{3V_{oph}I_{oph}}\right)$$

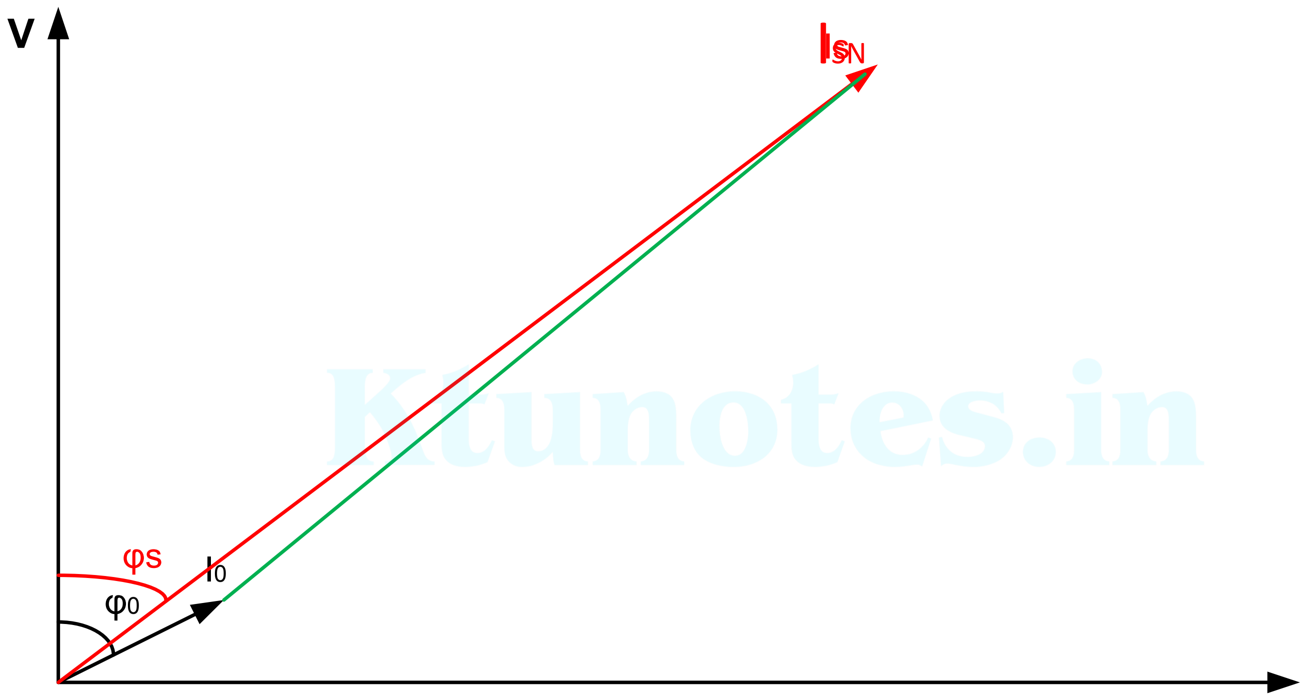
- $I_{0ph} = I_{0L}$

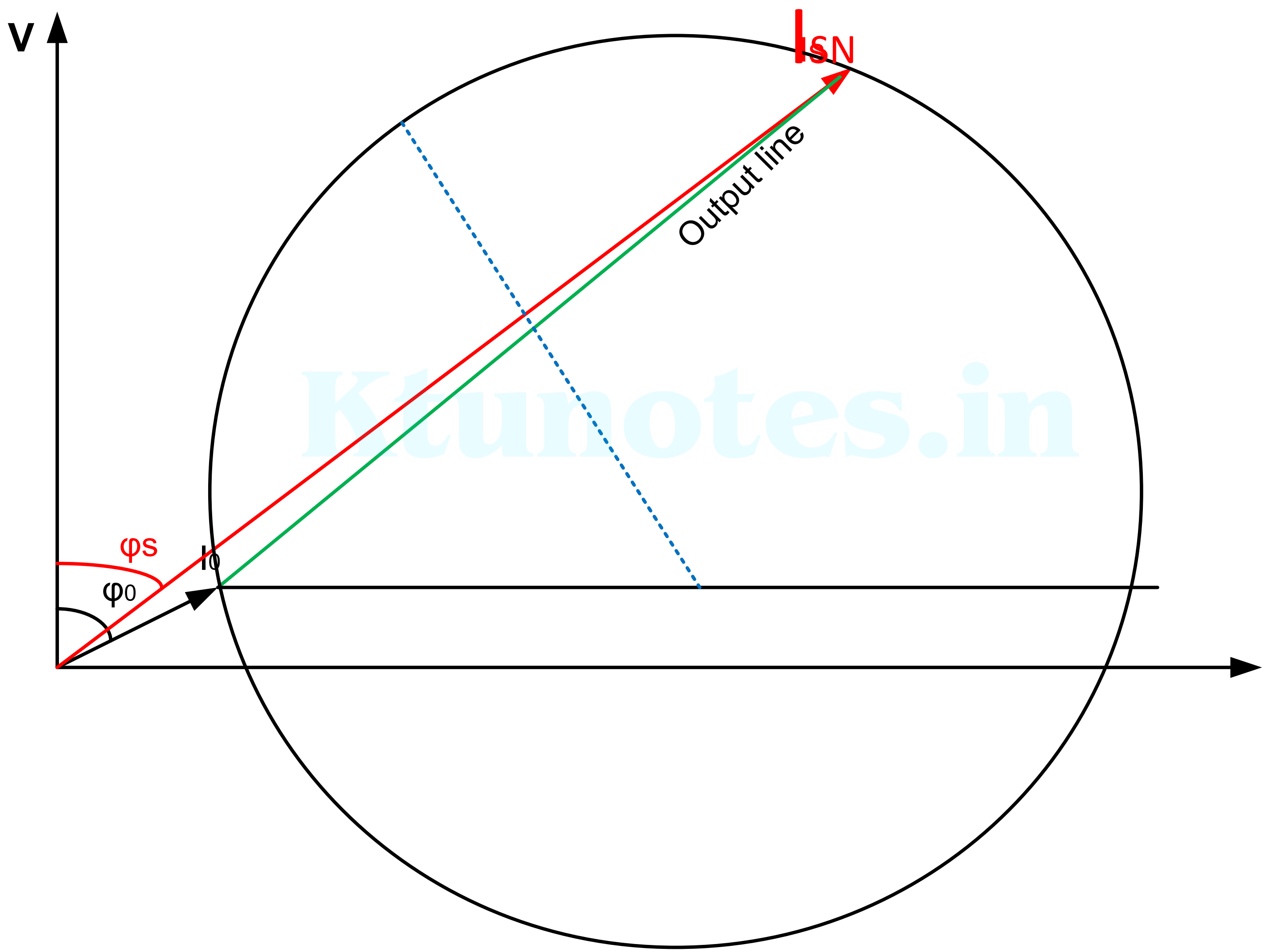
- From Blocked rotor test, calculate short circuit power factor angle (ϕ_{sc}) & short circuit stator current/phase (I_{SN}) when rated voltage is applied

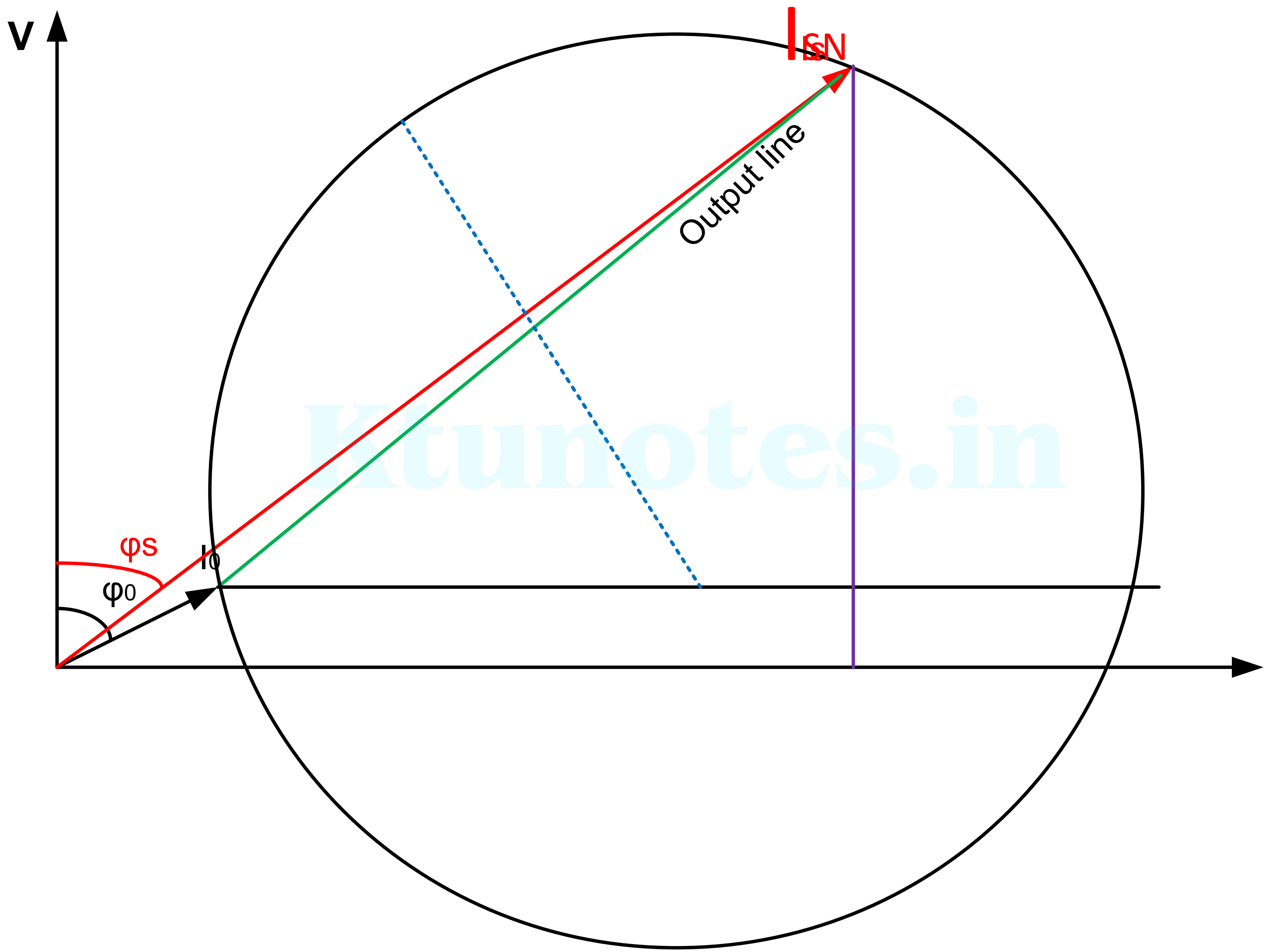
$$\phi_{sc} = Cos^{-1}\left(\frac{P_{sc}}{3V_{scph}I_{scph}}\right) \qquad I_{SN} = I_{scph}\frac{V_{Ratedph}}{V_{scph}}$$

Step 2 - Draw the horizontal axis OX & vertical axis OY
- The Voltage phasor V is taken as reference phasor & is drawn along Y-axis - With suitable scale (1cm = x A), draw phasor OO' with length corresponding to I_0 at an angle φ_0 from the vertical axis.

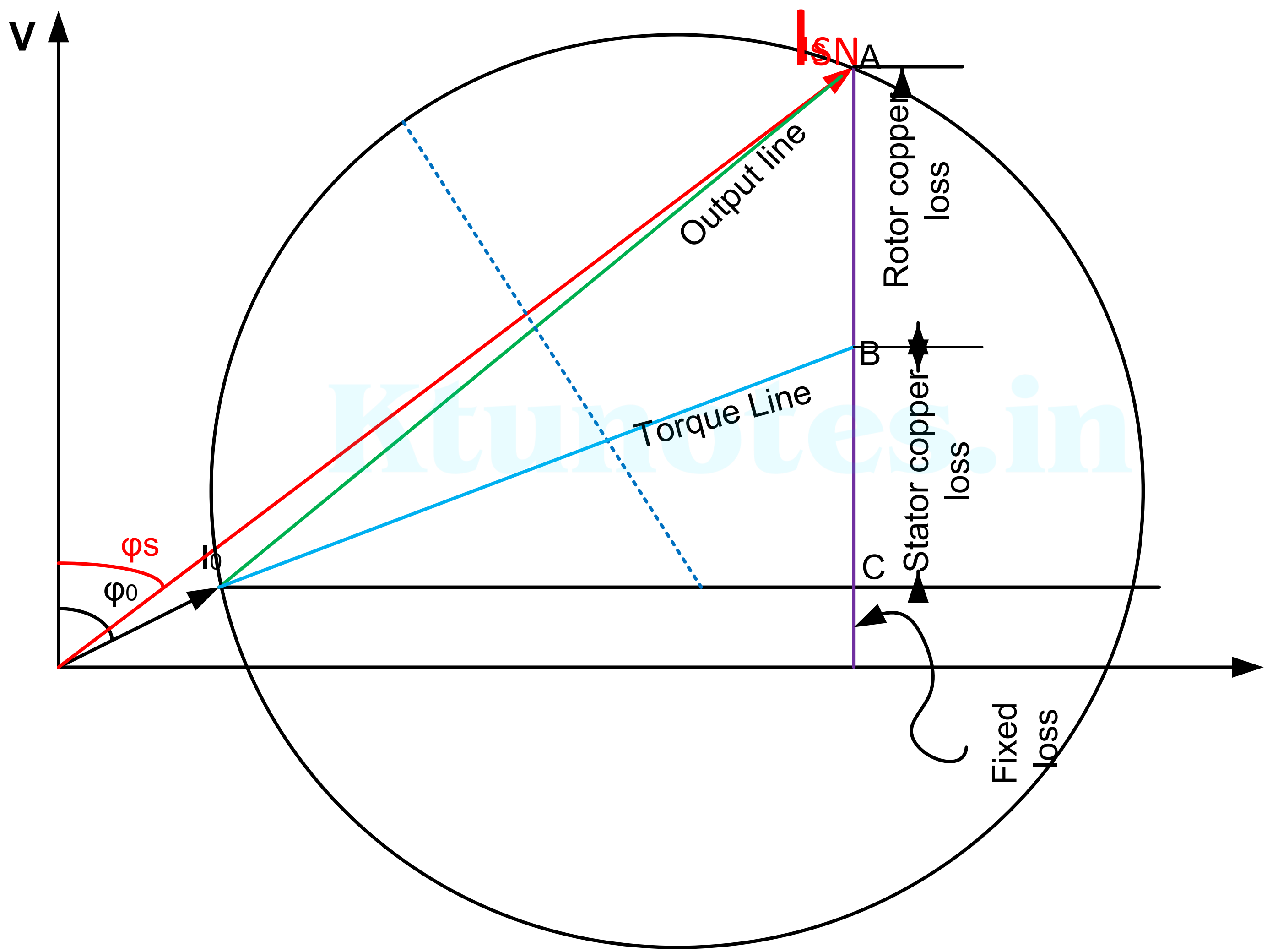

V
I_{SN}
I_s
φ_s
I_0
φ_0


V
φs
φ0
I0
Is
IsN

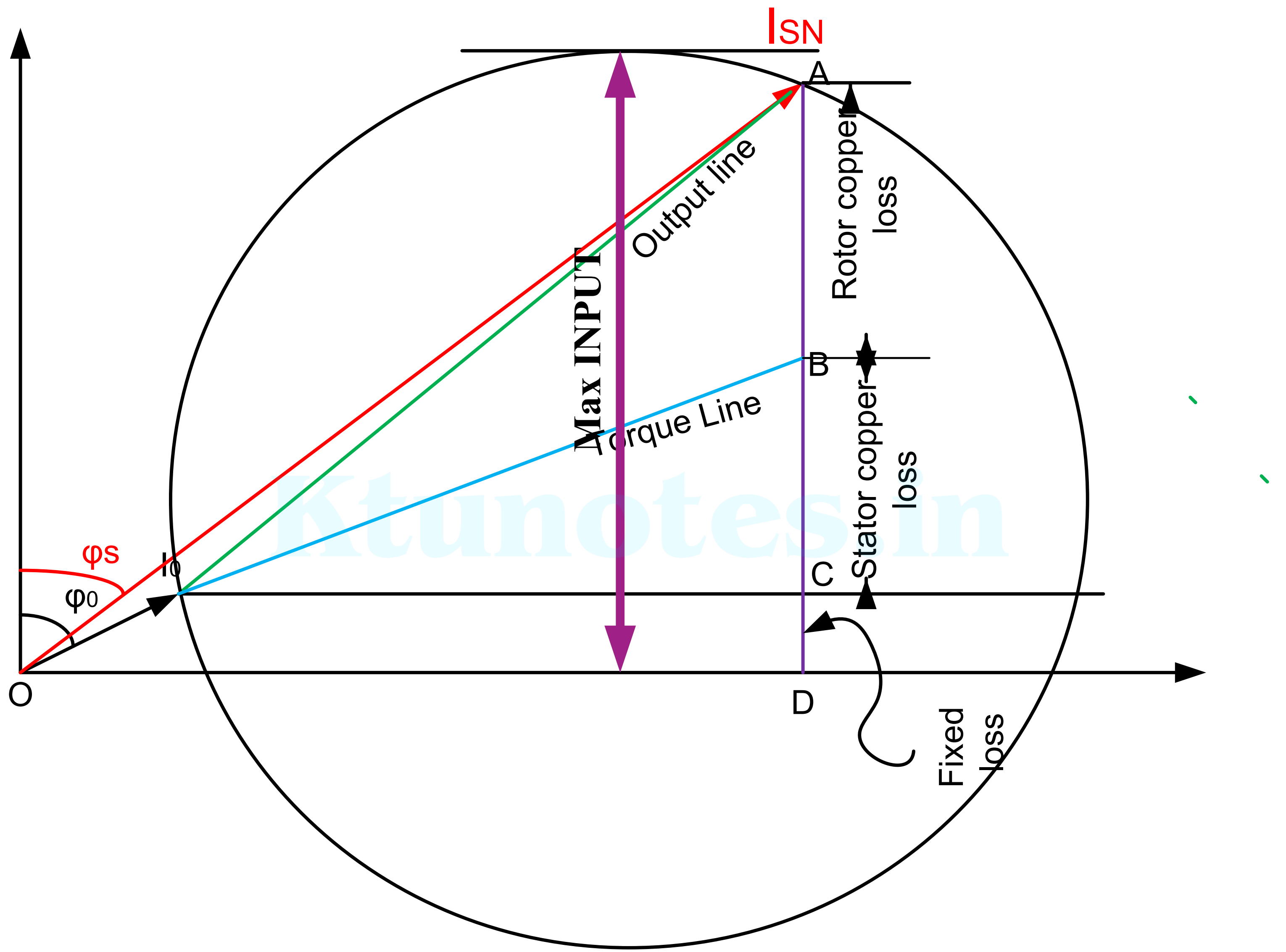

V
φs
φ0
I0
Output line
ISN



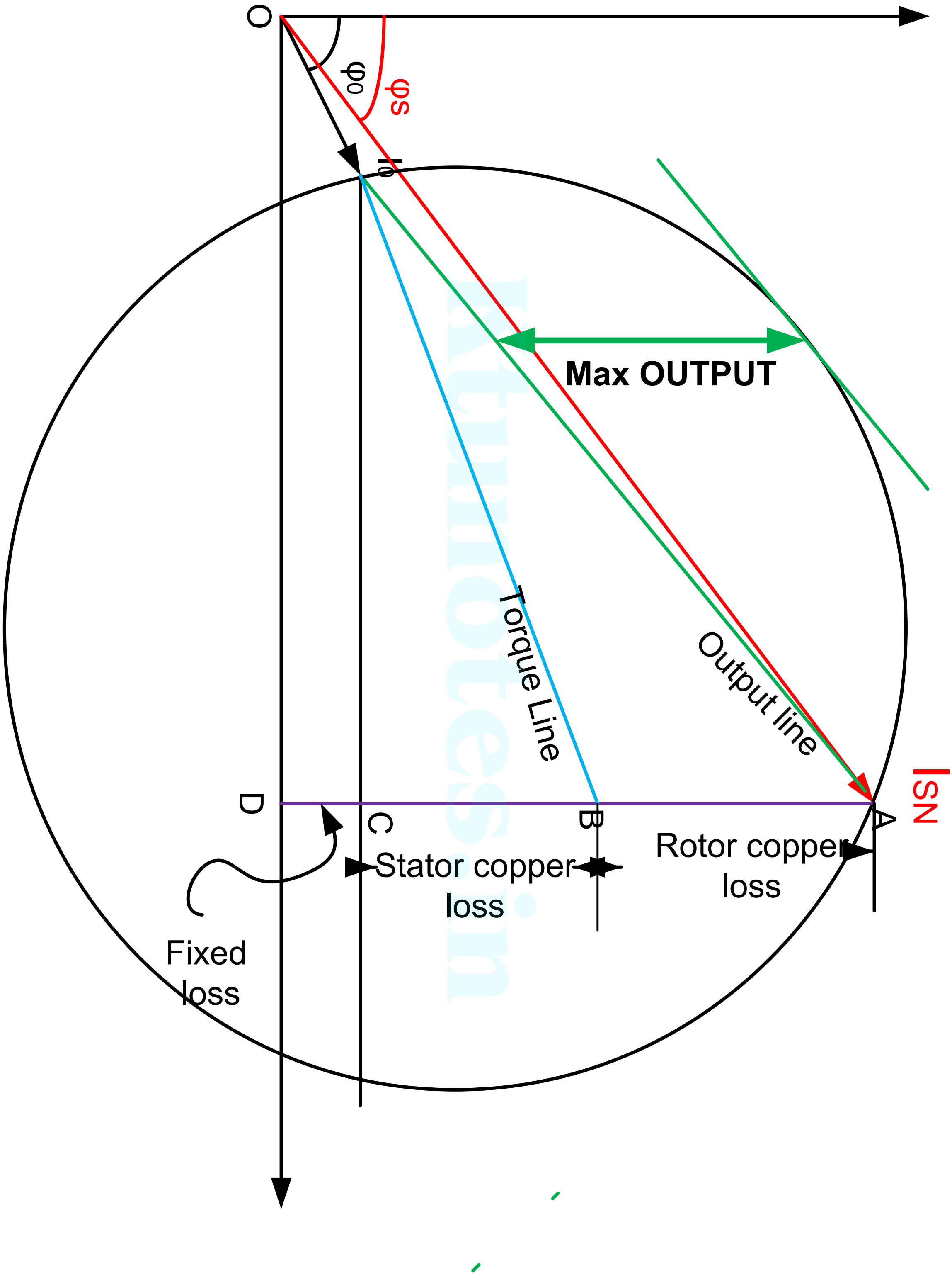
V
φs
φ0
I0
Output line
ISN

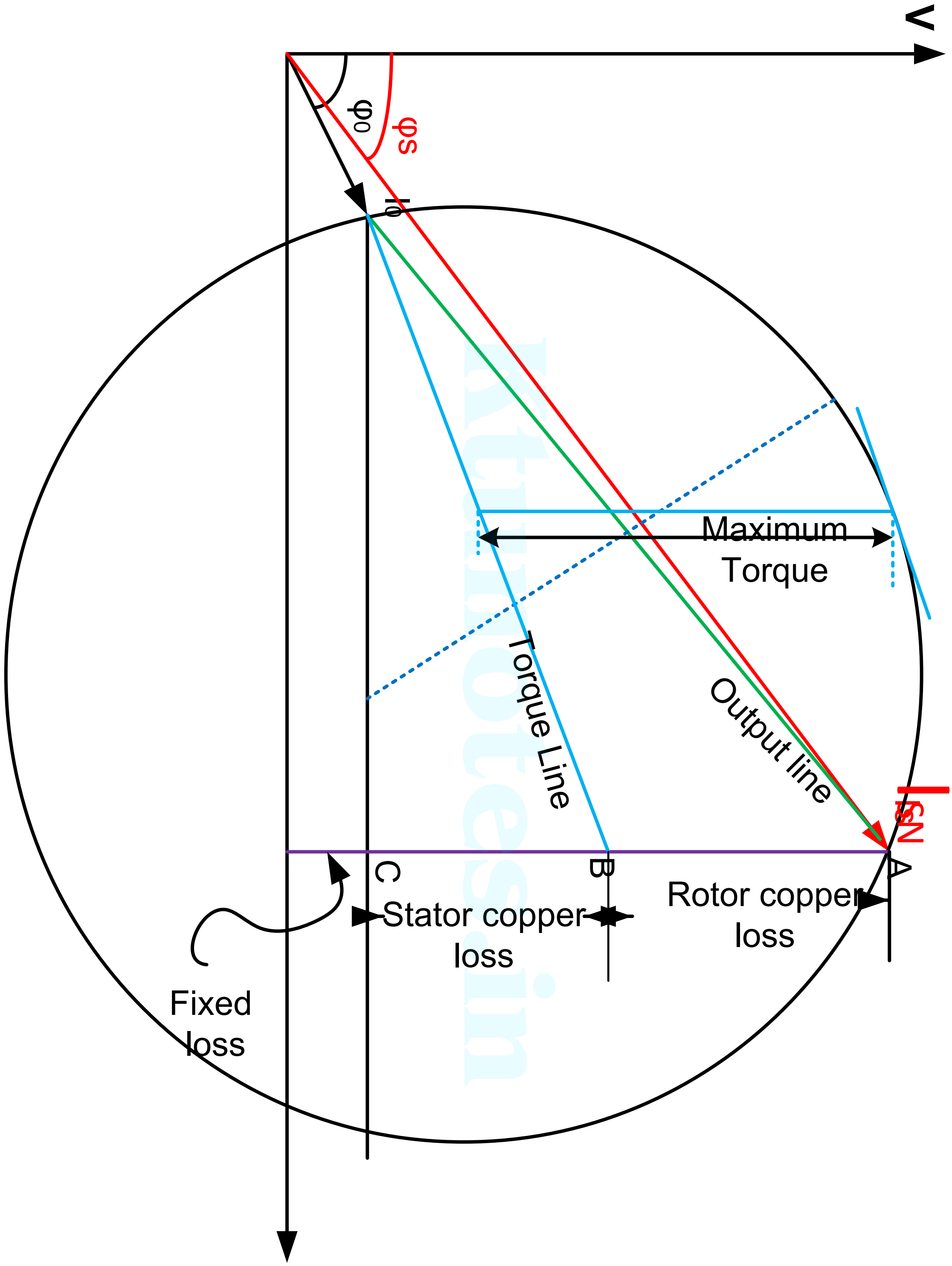


V
φs
φ0
I0
ISN
A
B
C
Output line
Torque Line
Rotor copper loss
Stator copper loss
Fixed loss

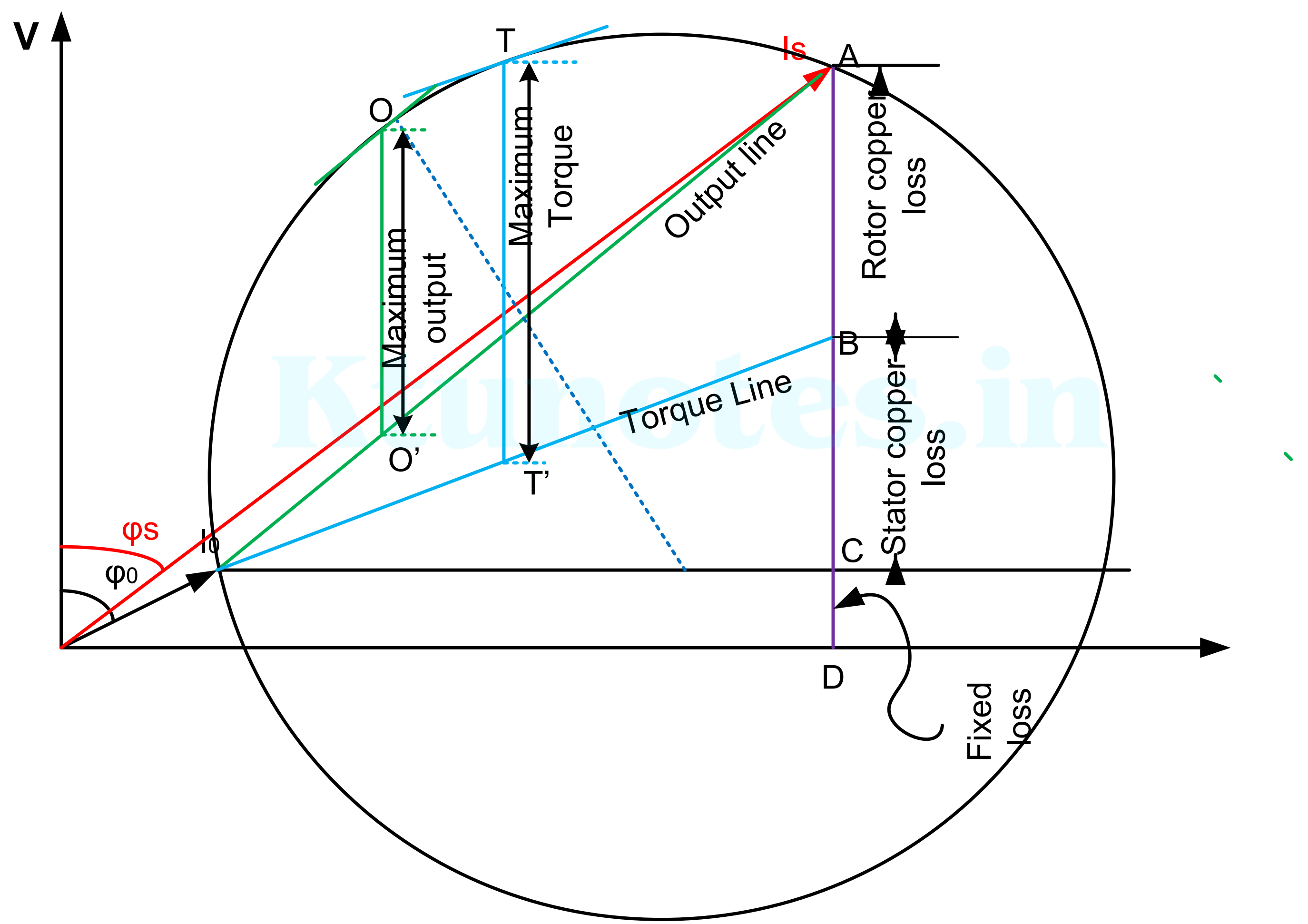


I_{SN}
A
Output line
Max INPUT
Rotor copper loss
B
Torque Line
Stator copper loss
φs
I_0
φ0
C
O
D
Fixed loss
Ktunotes.in

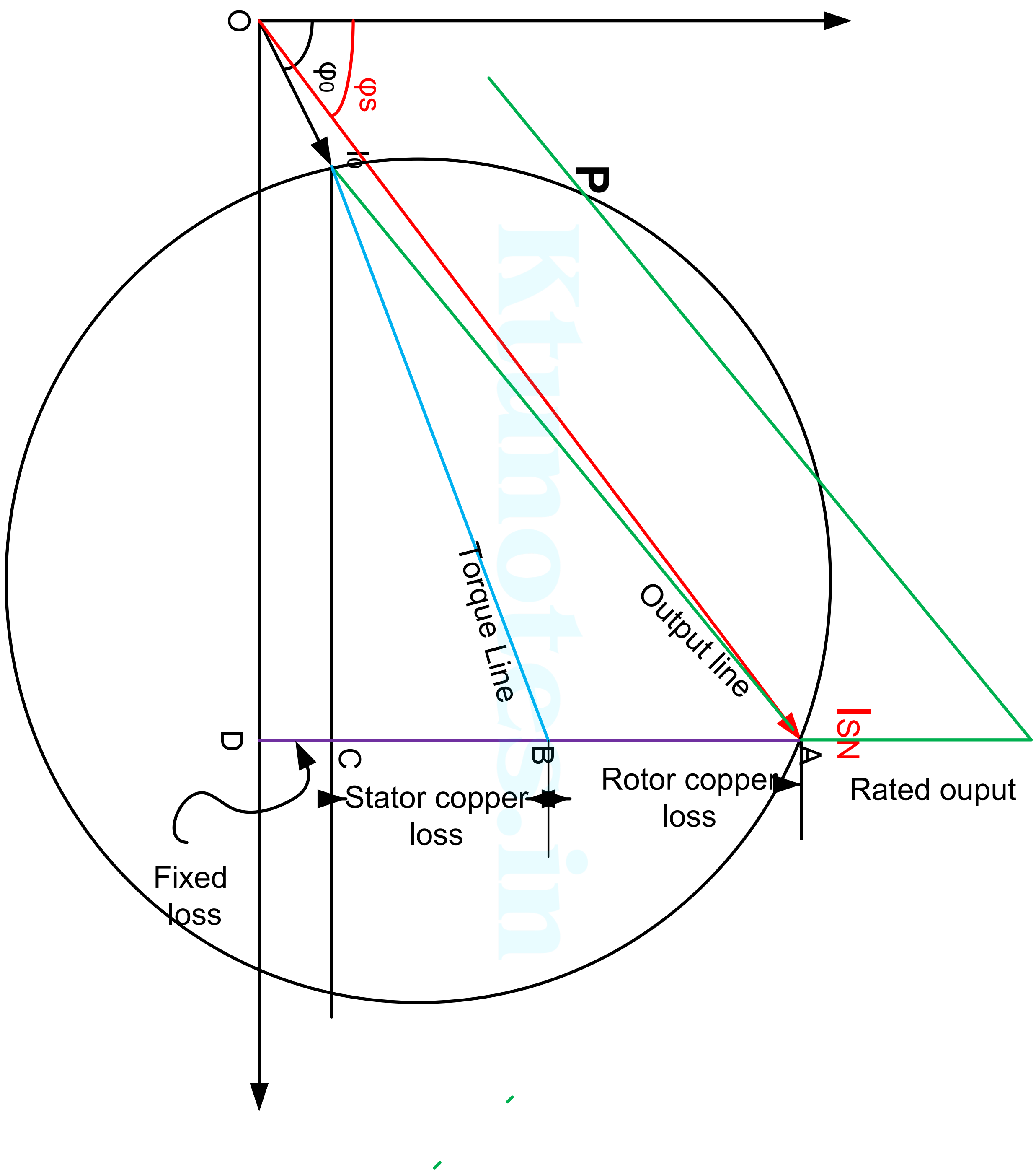

O
φ0
φs
I0
Max OUTPUT
Torque Line
Output line
ISN
D
C
B
A
Stator copper loss
Rotor copper loss
Fixed loss



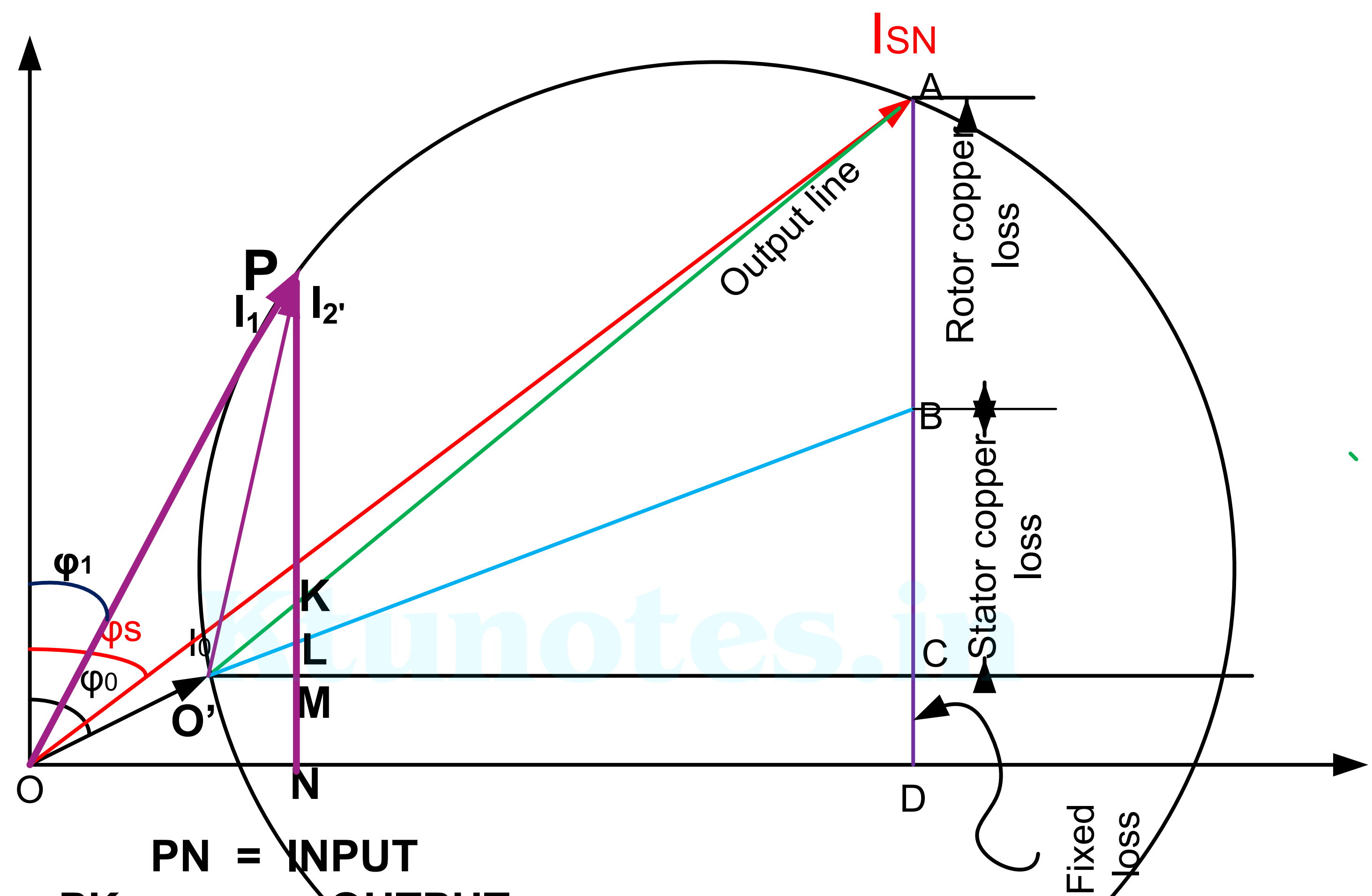
V
φ0
φs
I0
Maximum
Torque
Torque Line
Output line
IsN
C
B
A
Stator copper
loss
Rotor copper
loss
Fixed
loss



V
φs
φ0
I0
O
O'
T
T'
Is
A
B
C
D
Maximum output
Maximum Torque
Output line
Torque Line
Rotor copper loss
Stator copper loss
Fixed loss



O
φ0
φs
I0
P
Torque Line
Output line
ISN
D
C
B
A
Stator copper loss
Rotor copper loss
Rated ouput
Fixed loss



PN = INPUT

PK = OUTPUT

KL = Rotor copper loss

LM = Stator Copper loss

MN = Fixed loss

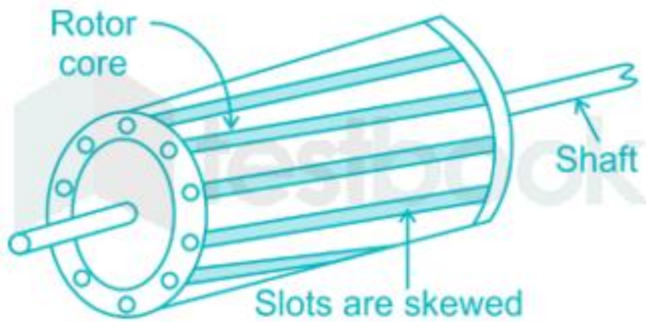
Efficiency= PK/PN

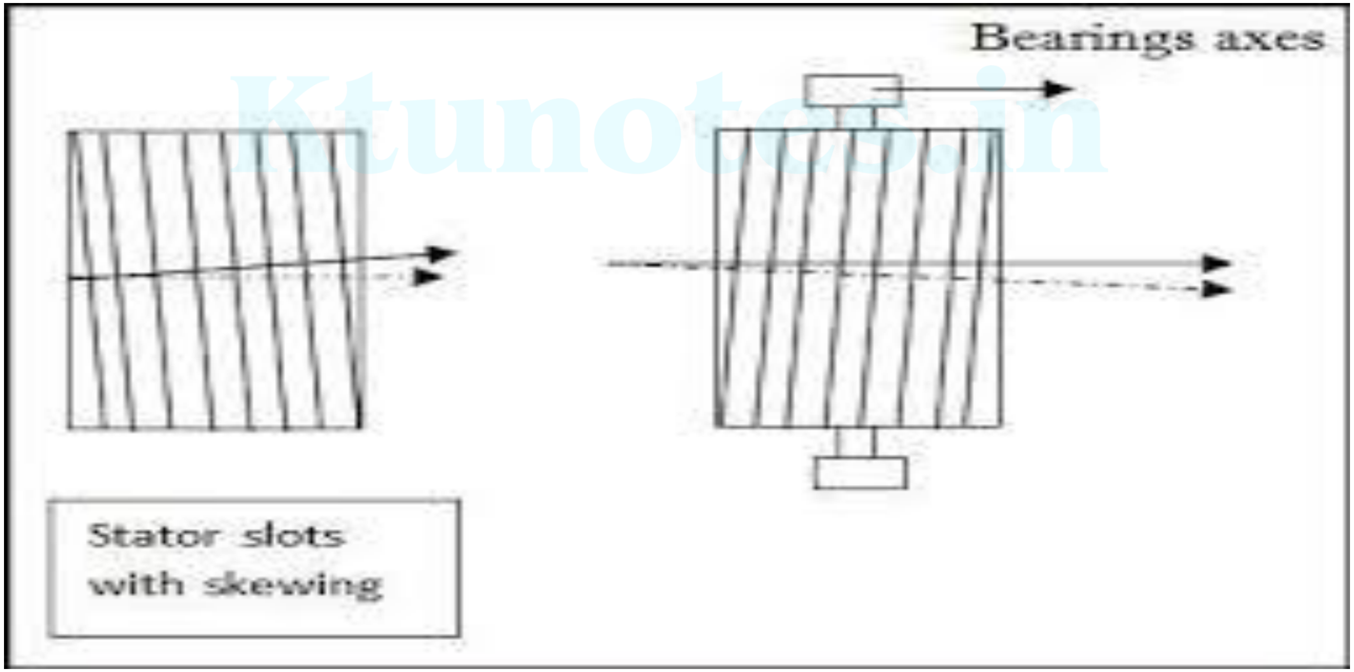
OP= Input Current

O'P= Rotor current reffered to stator

Cogging

With certain relationships between number of poles and of stator & rotor slots in cage rotor motors, peculiar behaviour may be observed when the machine is started.
If the number of slots on the stator & rotor are equal, machine may refuse to start at all. This phenomenon is known as Cogging.
It is due to the heavy magnetic pull between the stator & rotor teeth. Starting torque developed by the machine may be insufficient to overcome the radial magnetic pull. Hence the machine may not start.
Cogging is overcome by
1)having stator or rotor slots prime with respect to each other(ie, no common factor)
Example:
Number of stator slots = 24
Number of rotor slots = 41
2) Skewing rotor slots


Rotor
core
Shaft
Slots are skewed


Bearings axes
Stator slots
with skewing

Crawling

The tendency of the motor to run stably at a subnormal speed (eg. (1/7)th of normal speed) with a low pitched howling sound is called crawling.

In the torque-speed characteristics of induction motor(figure), reverse 5th harmonic torque & forward 7th harmonic torque are superimposed on the fundamental torque.

$(T_{developed} - T_{load}) \rightarrow$ accelerates the motor.

If motor developed torque follows fundamental torque curve, motor will accelerate to the point 'K' as a final operating point for the given load.

Any harmonic which causes a dip in the torque curve will cause motor to accelerate to point 'M' at which developed torque is equal to load torque.

The machine may continue to operate at point 'M' corresponding to a sub-normal operating speed.

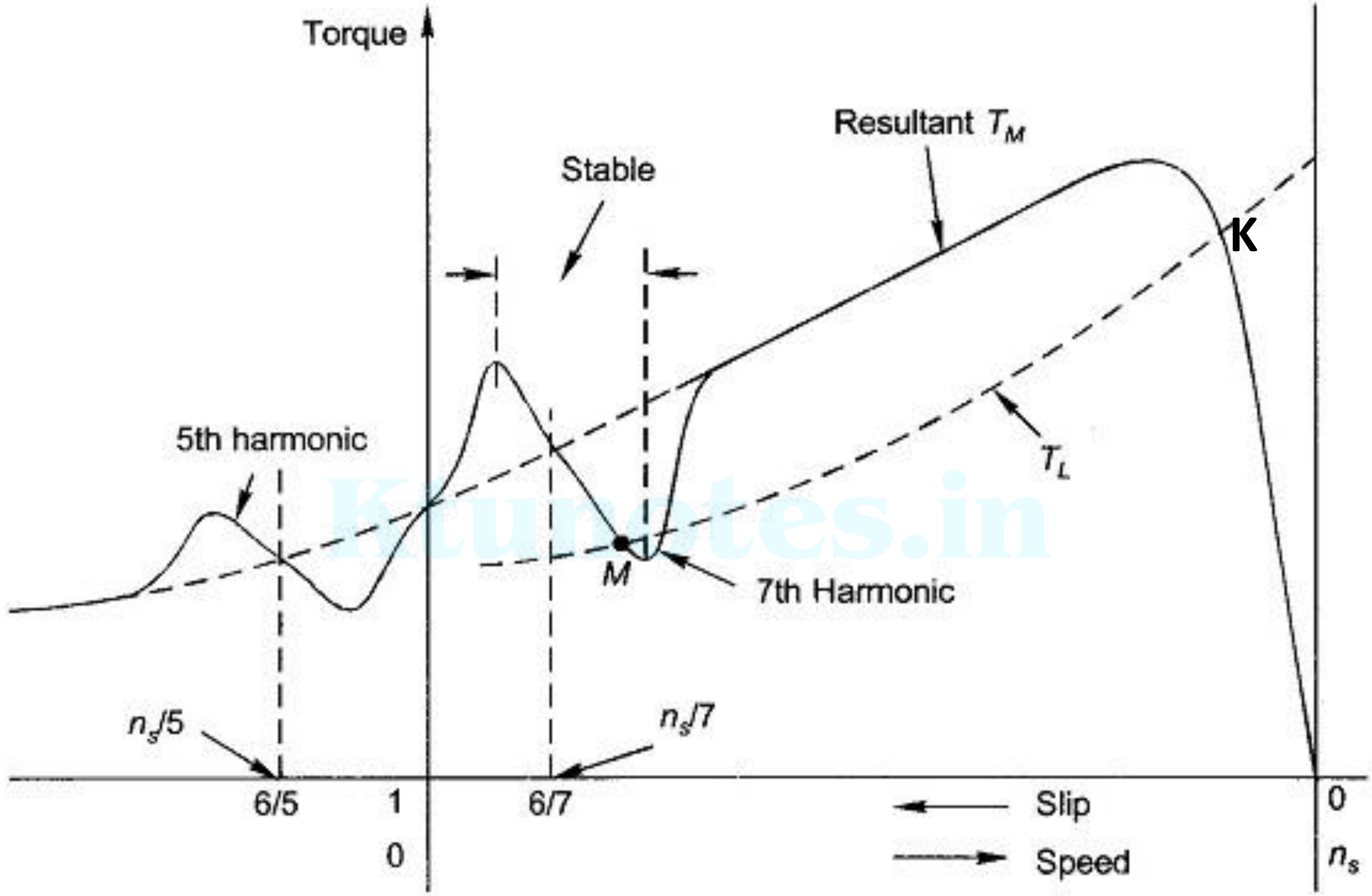


Fig. 9.37 *Torque-slip characteristic of a 3-phase induction motor showing the effect of harmonic asynchronous (induction) torques*

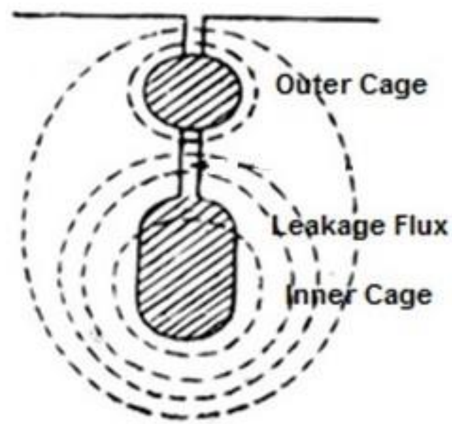
Proper choice of coil pitch & distribution of coils while designing the winding
→reduces harmonics in the air gap flux
→sinusoidal flux distribution
→ elimination of crawling.

High torque Cagemotors

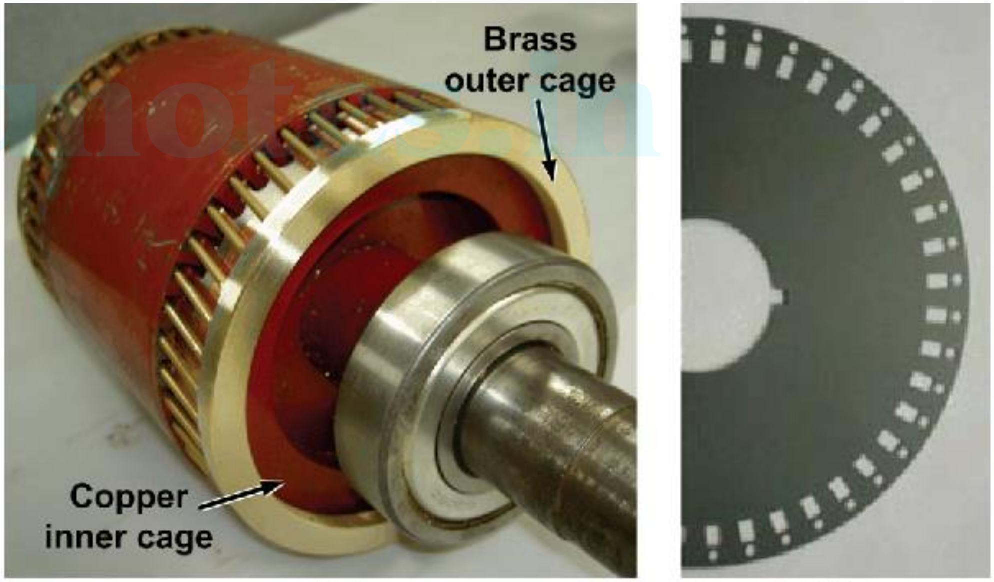
- SCIM has high efficiency, low cost, low maintenance etc.
- But it is having low starting torque, low p.f & high starting current
- A SRIM is having high starting torque, low starting current & high p.f due to high rotor resistance value
- So our requirement is high rotor resistance during starting to get more torque & low rotor resistance under normal operation
- It can be achieved by 2 methods
  - Double cage Induction motor
  - Deep bar Induction motor

Double cage Induction motor

- The stator remains same as normal SCIM.
- Rotor carry 2 set of squirrel cage winding
  - Inner cage winding – with low resistance & high reactance value made of Cu
  - Outer cage winding – with high resistance & low reactance value made of Manganese Bars



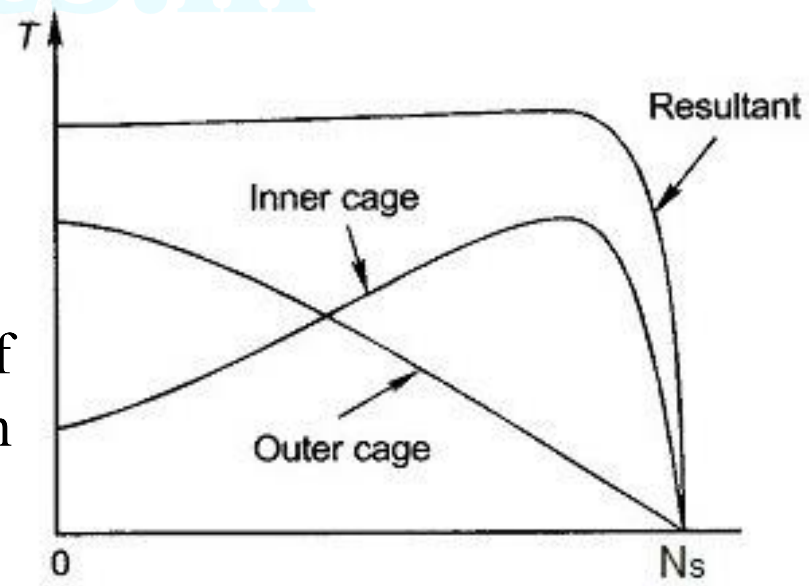
1. The view of end rings of the start-up cage and work cage.



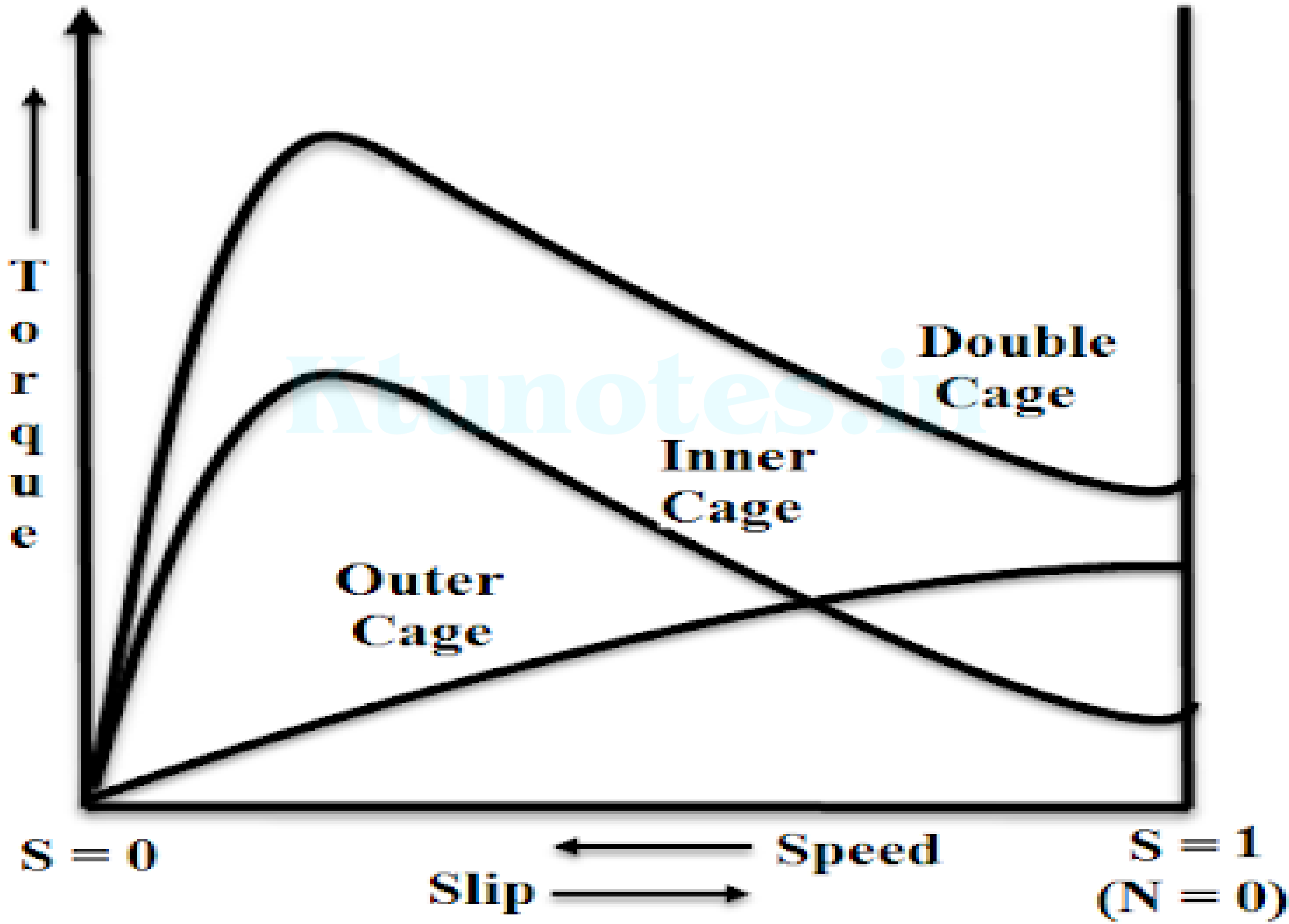
(a) (b)

'ig. 5 Rotor test. Custom designed separate end ring double cage

- Inner cage winding has higher cross sectional area than that of outer cage winding in order to have a resistance of $(1/4)^{th}$ of outer cage winding. End rings may be separate or combined
- During starting, $f_r = f$ & X_L of inner cage winding is high
- So impedance of outer cage winding is low compared to inner cage & most of current flows through outer high resistance cage to get more starting torque
- At normal speed, Slip decreases & reactance of inner cage decreases
- Now current divides between two windings according to their resistance value
- Because of low resistance, inner cage carry more current & produce desired running torque
- Torque – speed characteristics of individual cages & resultant are shown in graph

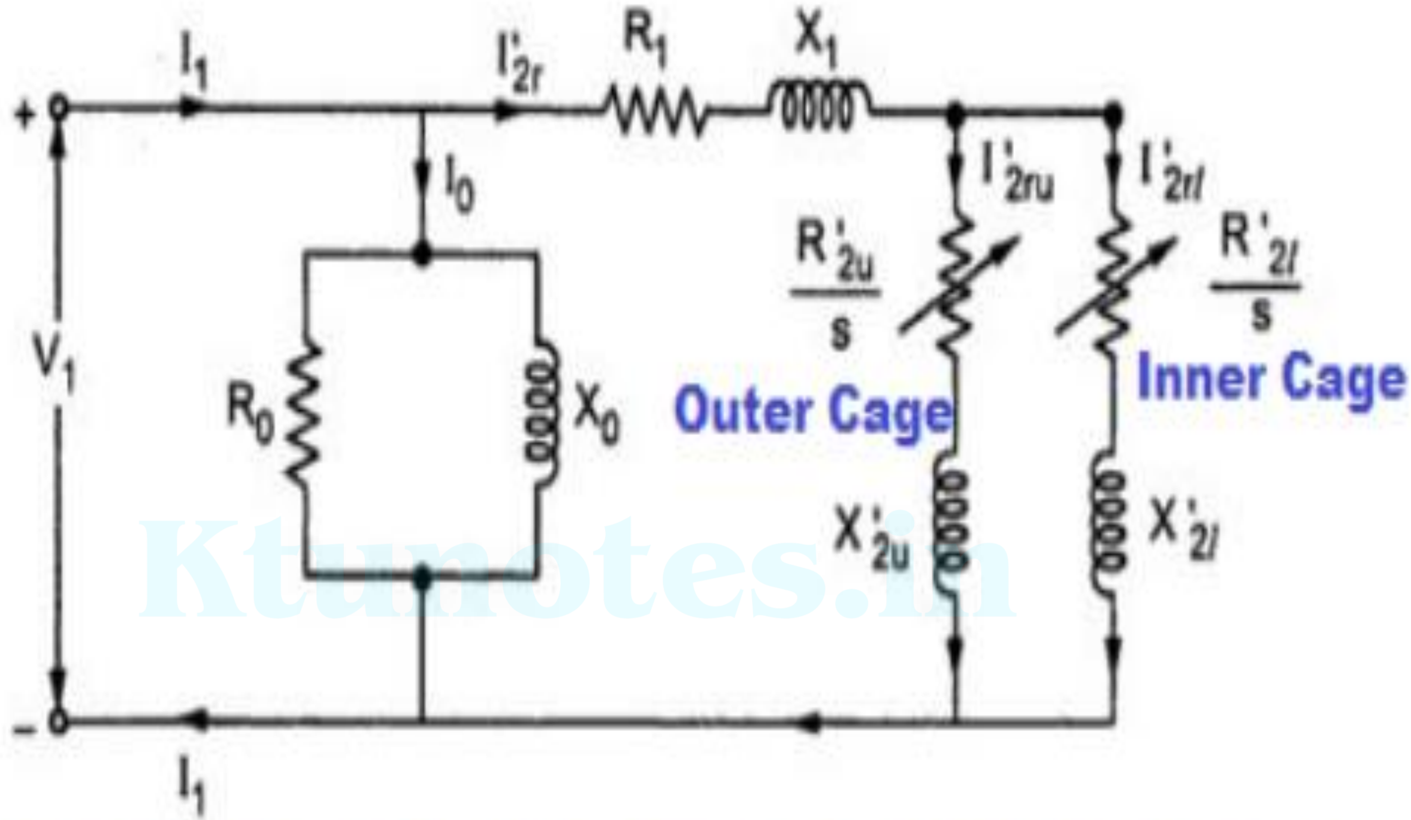


Torque-Slip Curve



Outer cage torque, inner cage torque & resultant torque.

- Desired T-N curve can be obtained by modifying individual cage resistances & leakage reactances.
- To change resistance → change areas of cross-section of bars.
- To change leakage reactance → change width of slot opening & depth of inner cage.
- Pull out torque of double cage induction motor is less than that ofnormal squirrel cage induction motor because, In double cage induction motor, two cages produce maximum torque at different speeds.
- Additional reactance of inner cage reduces full load power factor.



Equivalent Circuit of Double Cage Induction Motor

Need of starter

- In the case of an Induction motor at starting, when motor is standstill, the squirrel cage rotor is like a short circuited secondary of a transformer.
- Therefore if full supply voltage is applied during starting, current in rotor circuit will be very high & consequently stator also draw a high current from supply.
- Magnitude of this current depends on electrical design of motor & is usually 5 to 7 times rated full load current.
- To limit the current within safe value starter is used

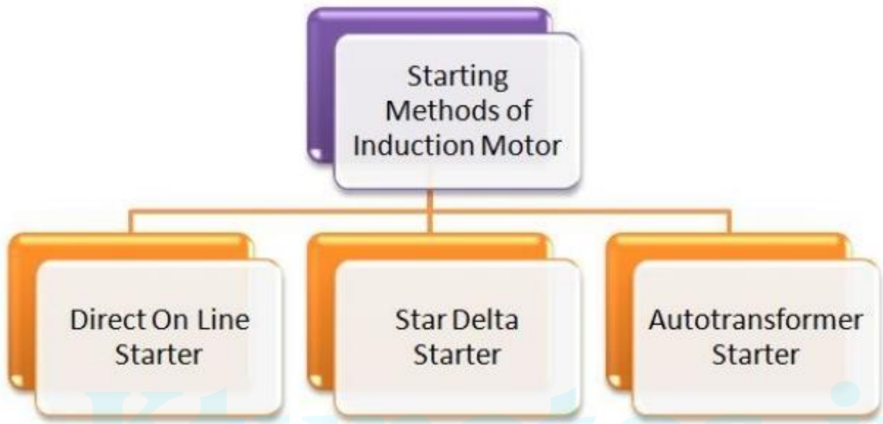
- **Functions of starter**

  -Limit the starting current within safe limit

  - Provide protection against overload & under voltage

1. Starting of Squirrel cage Induction motor

-Different type of starters used in SCIM are



1. DOL (Direct On Line) starter or Full voltage starter

- This is the most economical method of starting

- DOL starter is used based on following factors

* Power rating & design of motor
* Type of application
* Location of motor in distributed system
* Capacity of power system

- This method involves direct switching of poly phase SCIM to supply as shown in figure.

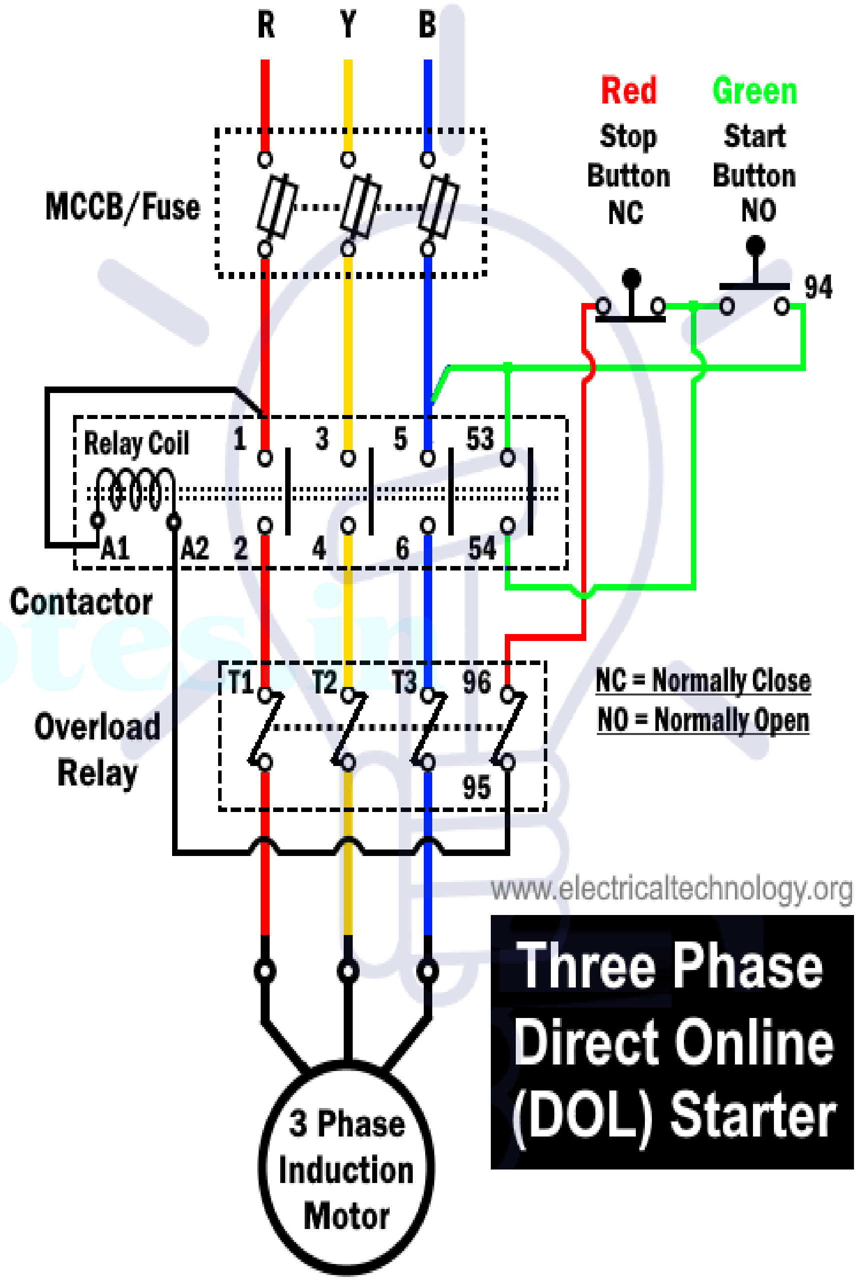
- It consist of start & stop button, a contactor, overload & under voltage protection devices.

- Start button is a normally open switch & Stop button is normally closed switch.

- For starting the motor, main switch is closed.

- Now start button is pressed & operating coil of contactor gets energized.

- Now the main contactor closes & connects the motor to supply.
- At the same time an auxiliary/maintaining contactor also closes & acts as a parallel circuit to start button. So now start button can be released.
- When stop button is pressed, operating coil get de-energized & all contactors get open & motor stops.



- If supply fails or supply voltage falls below certain value, main contactor & maintaining contactor gets open.
- When motor is overloaded, overload relay acts & operating coil gets de- energized
- Now motor is disconnected from supply by opening the contactors

- $I_{stline} = I_{st} = I_{sc}$ if stator is star connected.
- $I_{stline} = \sqrt{3}I_{st} = \sqrt{3}I_{sc}$ and $I_{st} = I_{sc}$ if stator is delta connected
- In DOL starter, Starting torque,

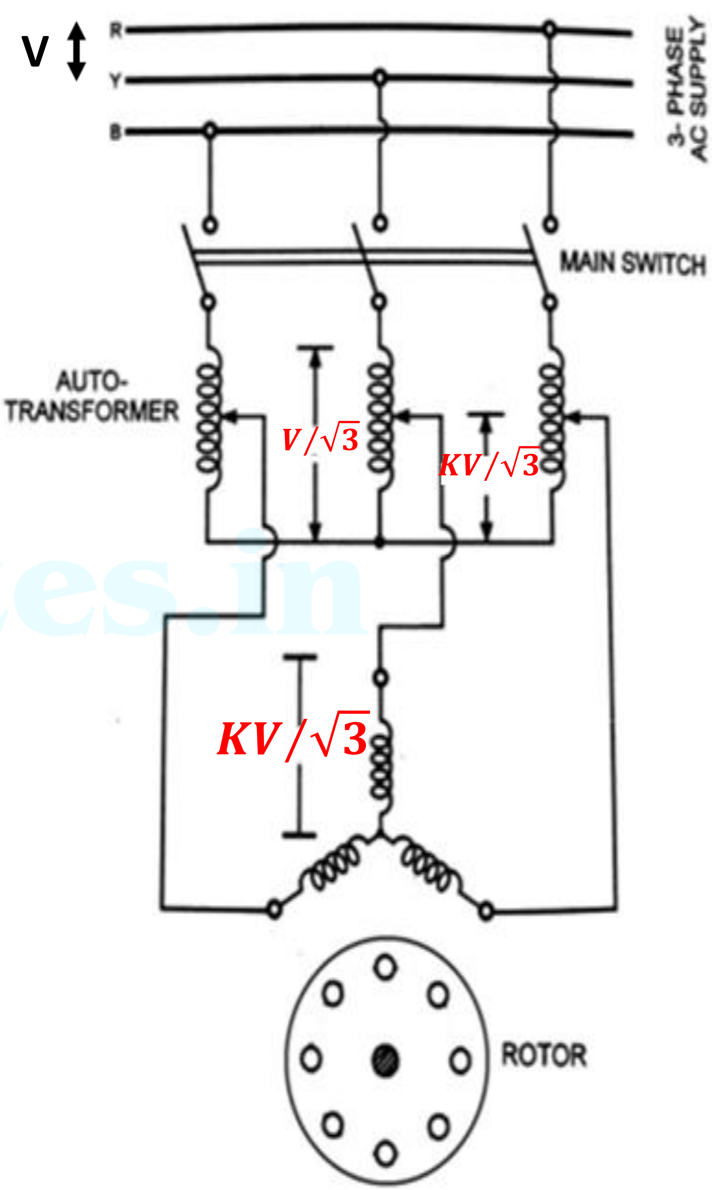
- $\frac{T_{st}}{T_f} = \frac{I_{st}}{I_f}^2 S_f$

$$T_{st} = T_f \left(\frac{I_{sc}}{I_f} \right)^2 S_f$$

2. Auto transformer starter

- Here the motor is supplied from an Auto transformer
- During starting, a low voltage is applied & as motor gains speed, auto transformer is adjusted to give rated voltage
- Auto transformer is adjusted manually or magnetically.
- Let the motor be started by an auto transformer having turns ratio 'k',then

- $I_{st} = \frac{KV/\sqrt{3}}{Z_{01}} = KI_{sc}$



- For an autotransformer, input power=output power

- $\frac{V}{\sqrt{3}} I_{stline} = \mathrm{K}\frac{V}{\sqrt{3}} I_{st} = \mathrm{K}\frac{V}{\sqrt{3}}.\mathrm{K}I_{sc}$

- $I_{stline} = K^2 I_{sc}$

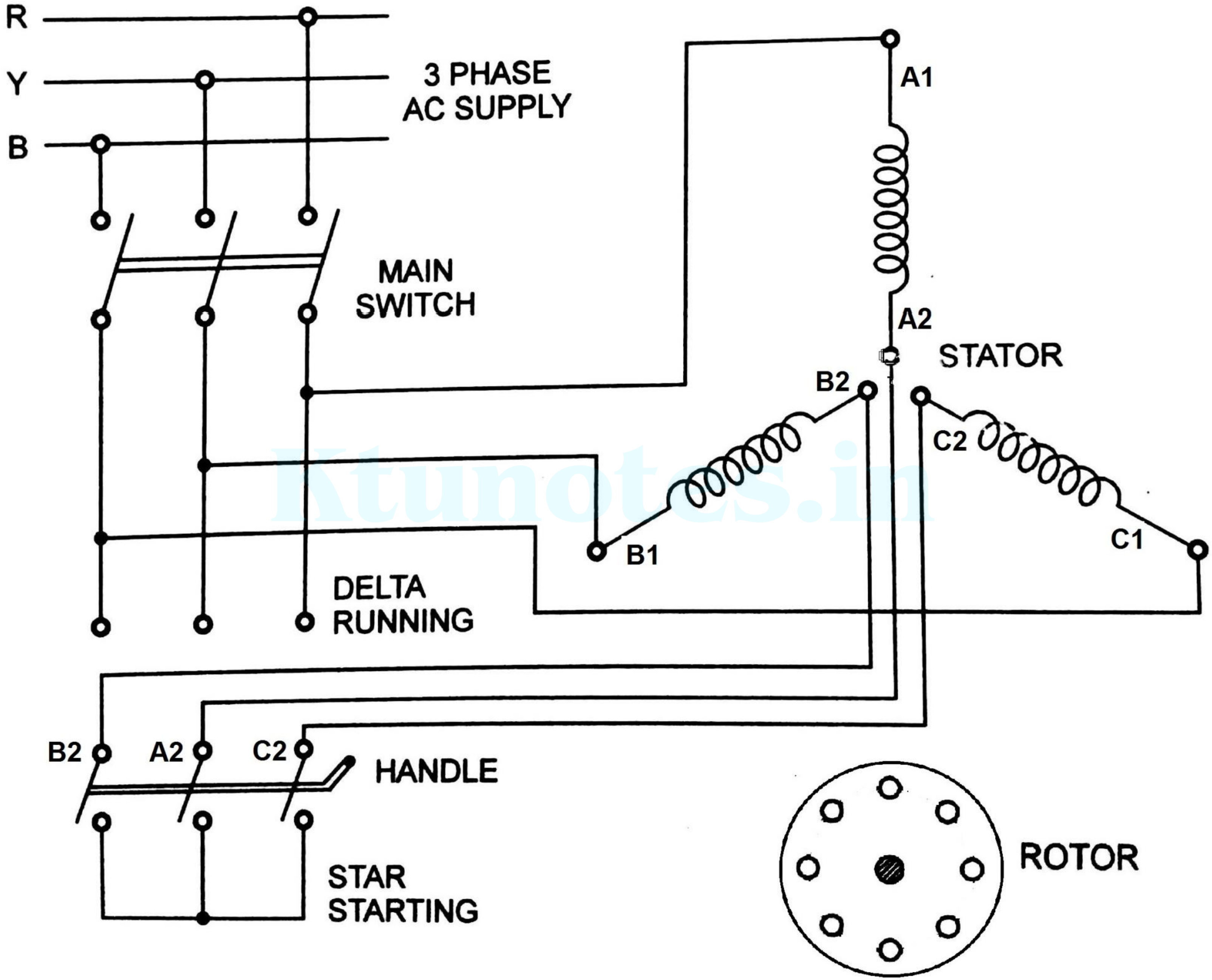
$$\frac{T_{st}}{T_f} = \frac{I_{st}}{I_f}^2 s_f = k^2 \frac{I_{sc}}{I_f}^2 s_f$$

$$T_{st} = T_f k^2 \left(\frac{I_{sc}}{I_f}\right)^2 S_f$$

- ❖ T_{st} with autotransformer starter = $k^2 T_{st}$ with direct starting
- ❖ I_{stline} with autotransformer starter = $k^2 I_{stline}$ with direct starting

3. Star – Delta starter

- This method of starting is based on the principle that, when 3 phase windings are connected in star, voltage across each winding is $(1/\sqrt{3})$ times line voltage
  - Where as when windings are delta connected, voltage across each winding is the full line voltage
- Here during starting, stator windings are star connected & when motor attains speed, same winding is connected in delta through a change over switch operated by a handle
- The connection diagram is shown in figure.
- Starting torque, $T_{st} = \frac{1}{3} T_f \left(\frac{I_{sc}}{I_f} \right)^2 S_f$ = (1/3) * Starting torque with DOL starter

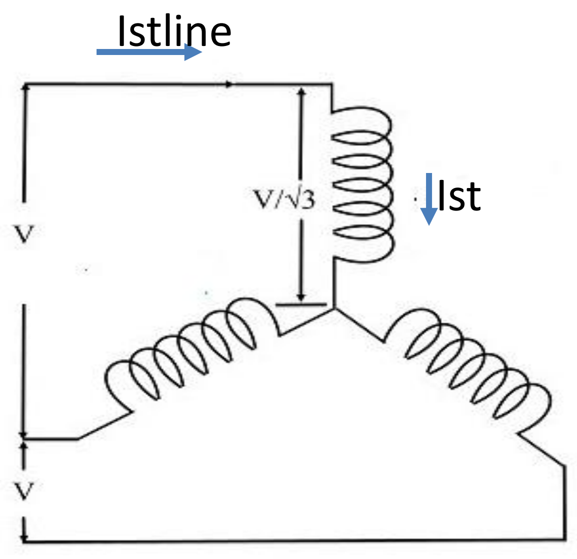

R
Y
B
3 PHASE
AC SUPPLY
MAIN
SWITCH
DELTA
RUNNING
A1
A2
STATOR
B2
B1
C2
C1
B2
A2
C2
HANDLE
STAR
STARTING
ROTOR

If started in star,

$$I_{st} = \frac{V/\sqrt{3}}{Z_{01}} = \frac{I_{sc}}{\sqrt{3}}$$

$$I_{stline} = I_{st} = \frac{I_{sc}}{\sqrt{3}}$$

$$\frac{T_{st}}{T_f} = \left(\frac{I_{st}}{I_f}\right)^2 s_f = \frac{1}{3}\left(\frac{I_{sc}}{I_f}\right)^2 s$$



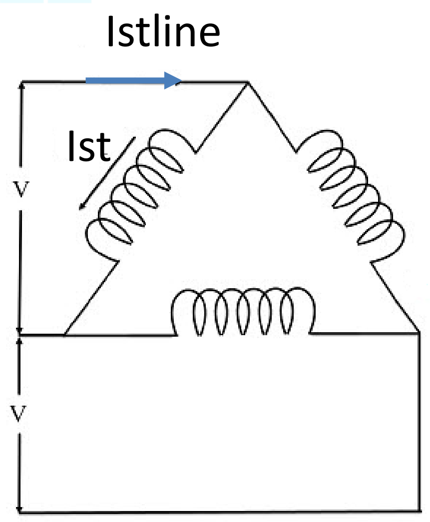
Star

If started in delta(Direct starting)

$$I_{st} = \frac{V}{Z_{01}} = I_{sc}$$

$$I_{stline} = \sqrt{3} I_{st} = \sqrt{3} I_{sc}$$

$$\frac{T_{st}}{T_f} = \left(\frac{I_{st}}{I_f}\right)^2 s_f = \left(\frac{I_{sc}}{I_f}\right)^2 s_f$$



Delta

$$\frac{I_{stline}\ with\ star-delta\ starter}{I_{stline}\ with\ direct\ starting} = \frac{I_{sc}/\sqrt{3}}{\sqrt{3}I_{sc}} = \frac{1}{3}$$

$$\frac{T_{st}\ with\ star-delta\ starter}{T_{st}\ with\ direct\ starting} = \frac{1}{3}$$

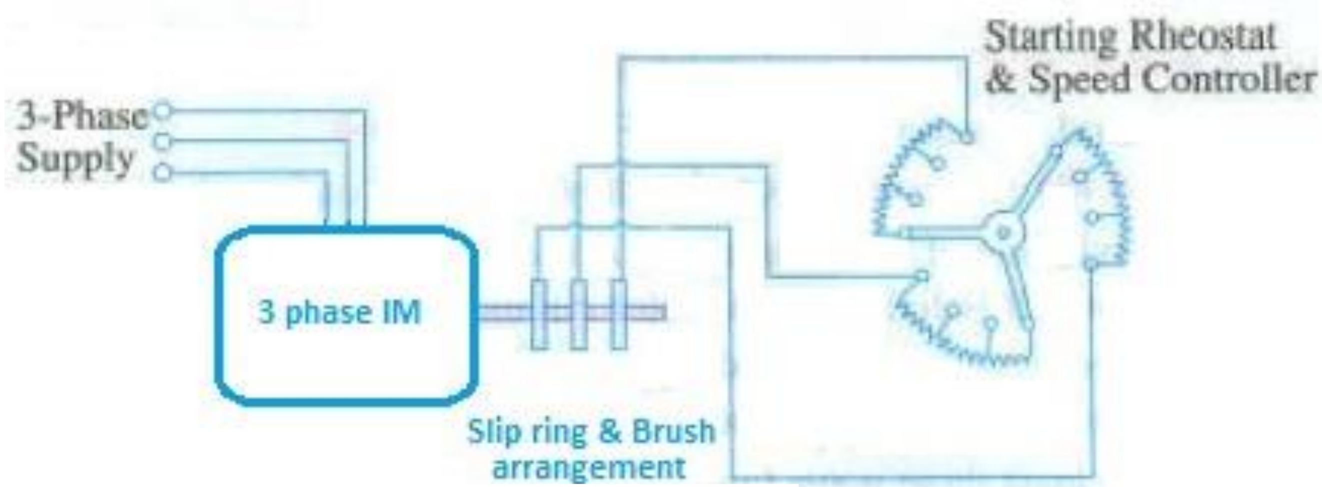
Star-delta starter is equivalent to autotransformer starter with K= $\frac{1}{\sqrt{3}} = 0.578$

Starting of SRIM

- The 3 types of starter already mentioned can be used in SRIM. But commonly used method is by using Rotor resistance starter

Rotor Resistance Starter

- SRIM is started by applying full rated voltage in stator & introducing variable resistance in each phase of rotor circuit
- The external resistance added in rotor circuit help to reduce starting current & also helps to improve starting torque
- During starting, rotor resistance is kept at maximum value & as motor gains speed, resistance value is decreased in steps & finally completely removed from circuit as motor attains rated speed
- The connection diagram for rotor resistance starter is shown in figure



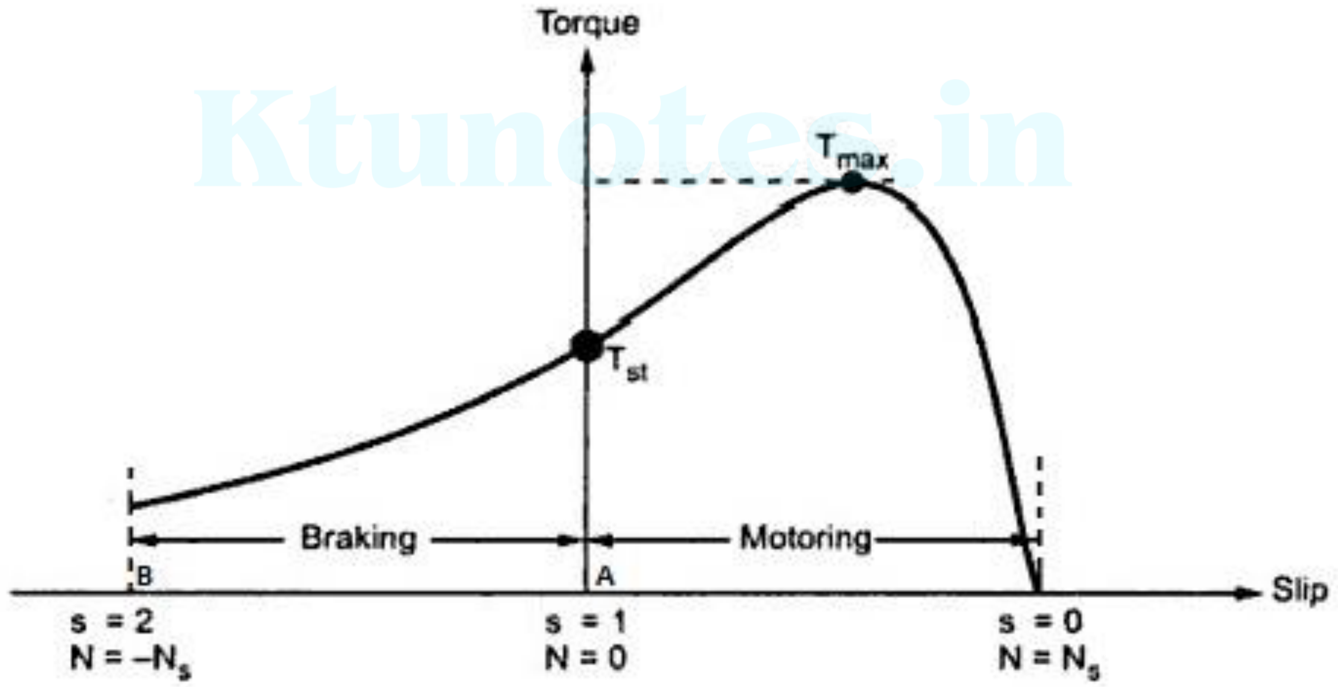
Braking of Induction motor

- The simplest method of stopping of an Induction motor is to disconnect the motor from supply mains.
- Now motor automatically stops due to combined effect of rotor & load (Kinetic energy in rotating parts gets dissipated as heat due to friction)
- When rapid stopping required, mechanical or electrical braking is employed
- There are mainly three methods of Electrical braking

1. *Plugging or Counter currentbraking*

- Plugging can be achieved in an Induction motor by interchanging connections of any two phases of stator w.r.to supply terminals
- This causes a reversal of the direction of rotational magnetic field & direction of torque produced by the motor.
- The torque produced by motor under braking condition acts as a braking torque & speed of motor decreases.
- During braking, motor runs in the opposite direction of rotational magnetic field & a large voltage induces in rotor windings. So rotor windings should be provided with additional insulation to withstand this high voltage

- During plugging, motor acts as a brake & absorbs kinetic energy from rotor causing its speed to fall.
- The associated power is dissipated as heat in motor.
- The condition for braking an induction motor can be studied by considering the Torque-Slip curve of the motor when extended beyond the point of slip = 1



- The y-coordinate at point B represents the torque at the instant of plugging & we can see that the torque increases gradually as motor speed approaches standstill
- *If supply line is not disconnected at zero speed, motor accelerates in the reverse direction*
- From T-S characteristics we can see that the braking torque is very small compared to maximum torque
- The braking torque & rotor current during braking are given by

$$T_b = \frac{KsR_2E_2^2}{R_2^2 + (sX_2)^2} \qquad I_2 = \frac{sE_2}{\sqrt{R_2^2 + s^2X_2^2}}$$

Advantage

- It is the quickest braking method

Drawbacks

- During braking power is wasted as heat & causes heating up of machine
- Due to large heat production, this braking can't apply frequently

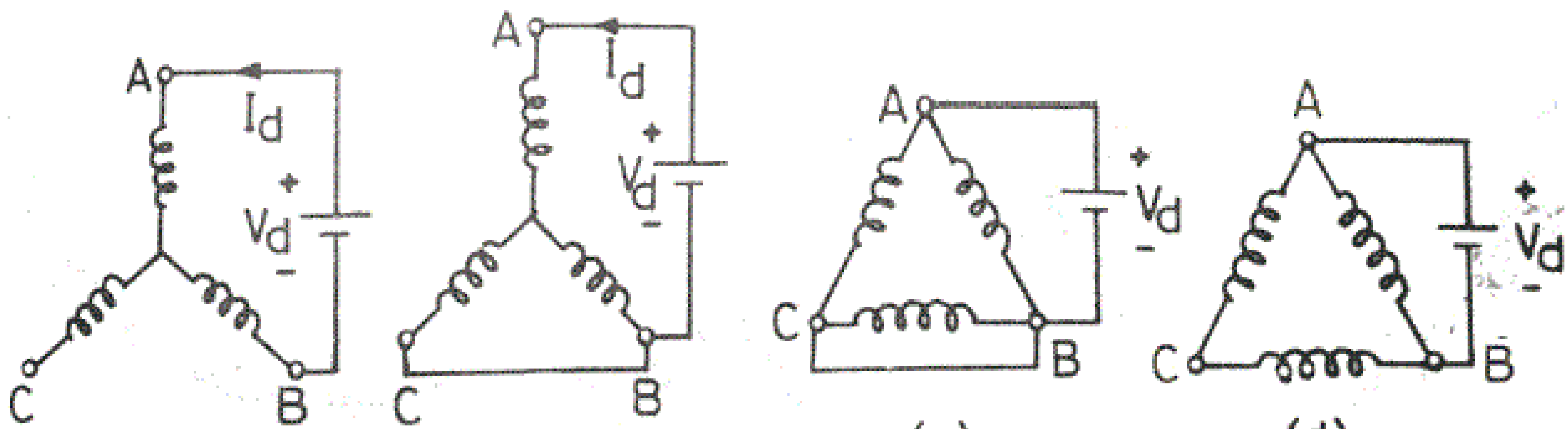
2. *Dynamic braking*

- Here during braking the stator windings are disconnected from AC supply & is connected to a DC source
- Now a stationary DC field is produced in stator & act as field excitation of a DC motor
- The rotor windings of Induction motor acts as armature winding
- When stator windings are disconnected from AC supply & excited with DC, the magnetic field produced will be stationary & rotor will be rotating
- Now current induces in rotor conductors due to generating action.
- The developed energy dissipates in the rotor resistance.
- Hence the motor speed reduces.

- Various methods for connecting the stator windings to DC source are shown in figure
- The DC excitation can be given either by using an independent DC source or from a rectifier

Advantages

- Less heat is produced as compared to plugging
- Provides smooth braking
- Here there is no tendency of machine to run backwards

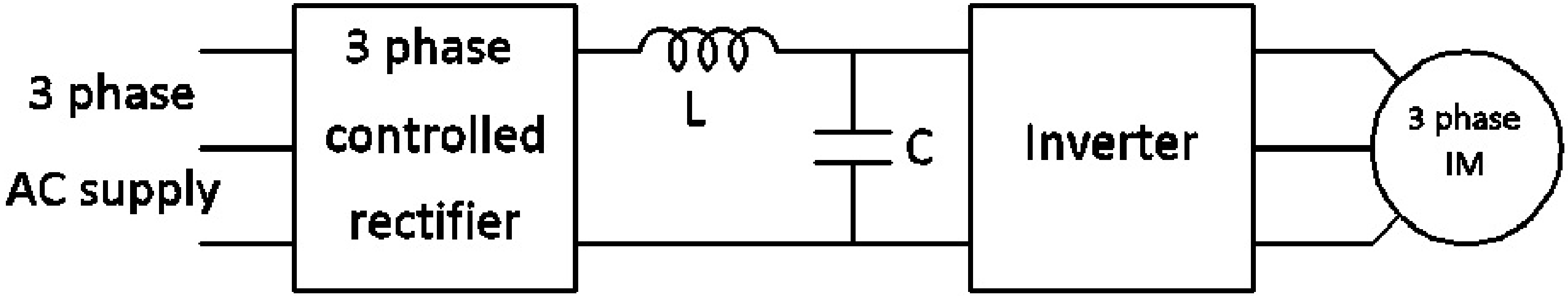


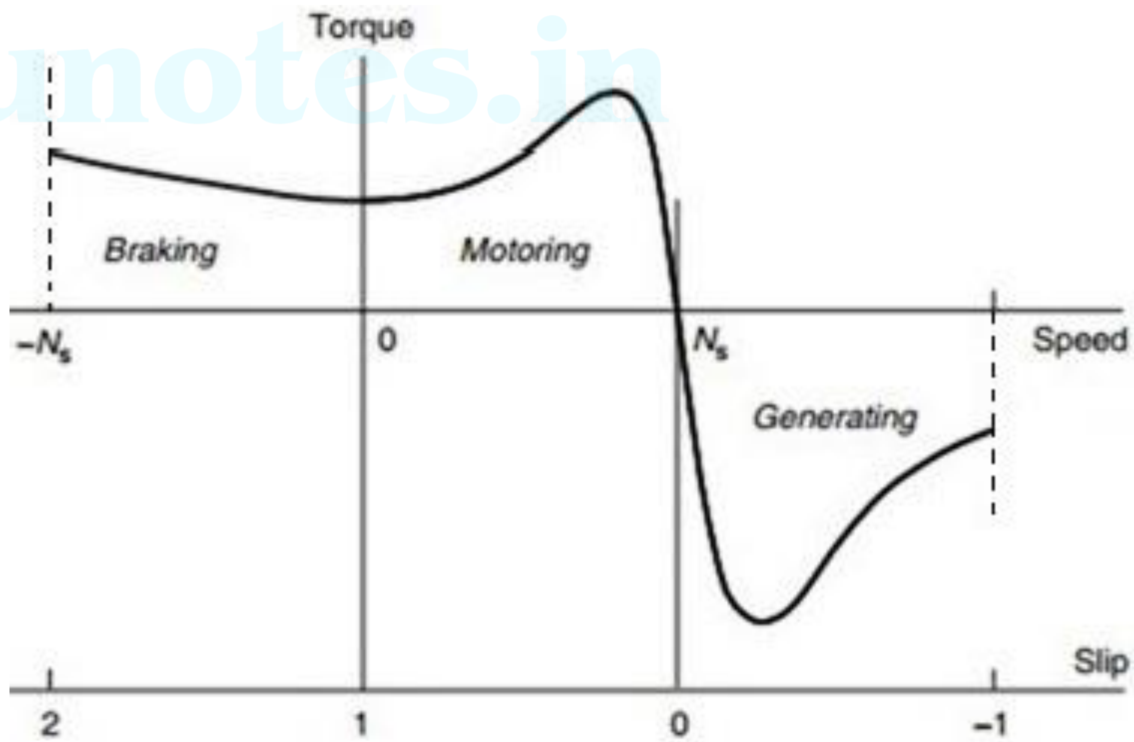
(a) (b) (c) (d)

3. *Regenerative Braking*

- During regenerative braking, Induction motor works as a generator & electrical power generated is fed back to the supply
- This generator action produces the required braking torque
- When speed of an Induction motor increases above synchronous speed, it acts as an Induction generator
- The Induction motor can be made to operate at speed above synchronous speed by any of the following method
  - Switching over to a low frequency supply in frequency controlled induction motor drives
  - Switching over to a large number of poles operation from a smaller one in multi speed squirrel cage motors
- During braking, slip & torque become negative & machine acts as a generator receiving mechanical energy & giving back electrical energy to supply
- The torque – speed characteristics of an Induction motor under regenerative braking operation is shown in figure

- The block diagram of an Induction motor drive equipped with regenerative braking is shown in figure.





Advantage

- The generated power is fully used

Drawback

- The motor should be supplied from a variable frequency supply

Speed control of Induction motor

- 3 phase Induction motors are practically a constant speed motor like a DC shunt motor
- A DC shunt motor can be made to run at any speed within limits to obtain required performance by armature & field control
  - But in Induction motor speed control, it is not that easy
- In Induction motor, as speed changes, efficiency & performance changes
- In an Induction motor, speed is given by $N = N_S (1-S)$

$$\text{i.e, } N = (120f/p)*(1-S)$$

- Basic methods of speed control of induction motor are

i) By changing the number of poles

ii) By varying the supply frequency

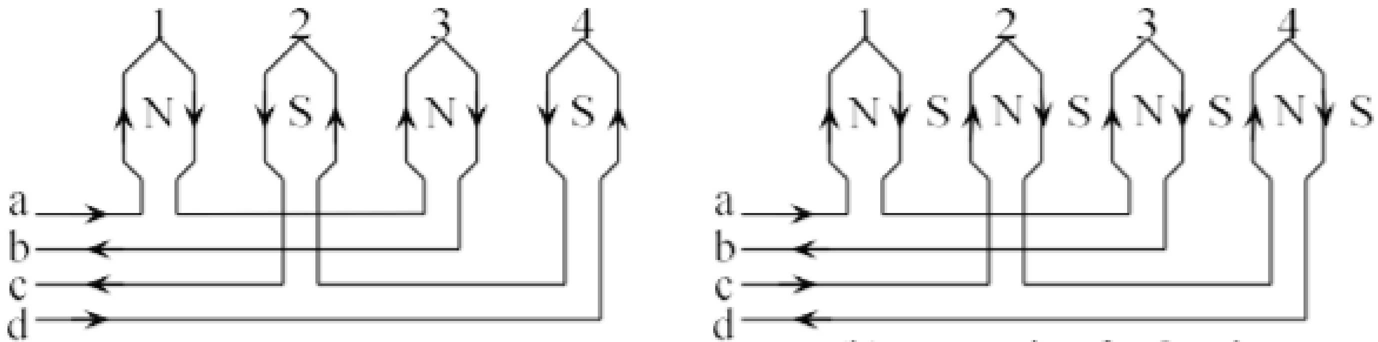
iii) By varying the stator voltage

iv) By varying the rotor resistance

Method (i) is applicable only for squirrel cage induction motors and (iv) only for slip ring induction motors.

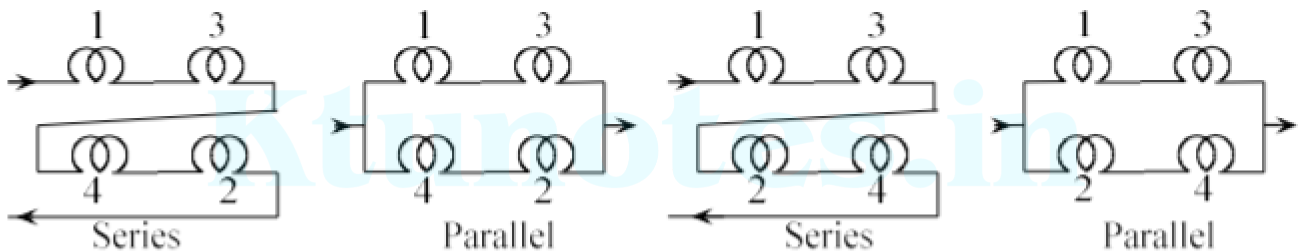
i)POLE CHANGING

- For a given frequency, the synchronous speed is inversely proportional to the number of poles.
- Synchronous speed and hence the motor speed can be changed by changing the number of poles.
- this method of speed control is only used for squirrel cage induction motors.
- If an induction motor is to run at different speeds, one way is to have different stator windings for the motor so that it will have different synchronous speeds and the running speeds. But this method is expensive.
- Another method is to use one winding but with suitable connections for a change-over to double/halve the number of poles. The stator winding per phase is divided into two coil groups. By reversing the current in one coil group, change in number of poles by a factor 2 can be achieved. The two coil groups can be connected in series or parallel.



(a) connection for 4 poles

(b) connection for 8 poles

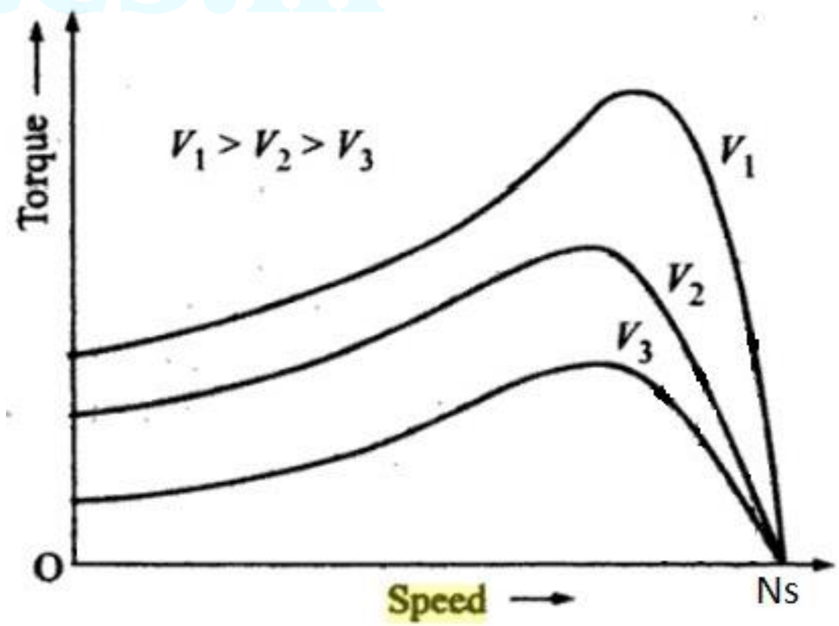


ii) Stator voltage control

- Here the slip is controlled to control the speed of motor
- Motor is supplied from a variable voltage constant frequency supply
- In an Induction motor, $T \alpha E_2^2$ & $E_2 \alpha V$, supply voltage

  e, $T \alpha V^2$

- So by varying stator voltage, T & hence slip can be controlled
- T – Slip characteristics of an Induction motor with variation in supply voltage is shown in figure



- Applied voltage can be varied by using a 3 phase autotransformer or tap changing transformer

Advantage

- Simple, low cost

Drawbacks

- Voltage can’t be increased above rated value. So speed control below rated speed is only possible
- A large change in voltage is required to get a small change in speed
- The torque developed changes with change in supply voltage

iii) Rotor resistance control

- This method is applicable to SRIM alone
- Here slip is varied to vary speed of motor

$$T = \frac{KsR_2E_2^2}{R_2^2 + (sX_2)^2}$$

Since, during running condition, $s^2X_2^2 << R_2^2$

$$T \propto \frac{sV^2R_2}{R_2^2} \propto \frac{sV^2}{R_2}$$

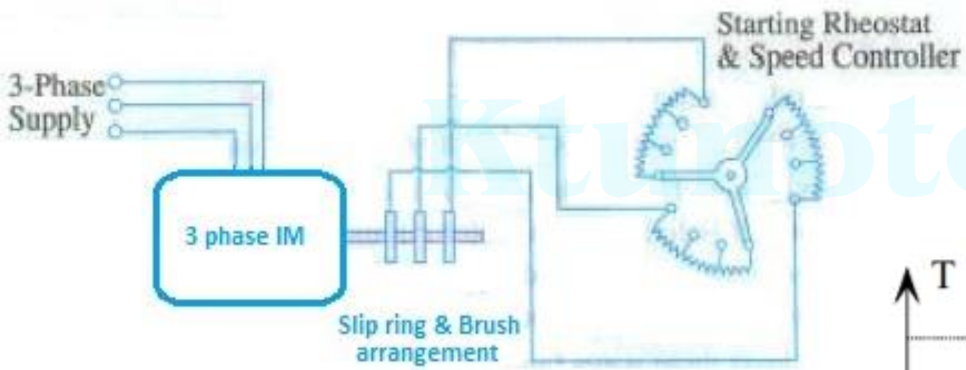
If supply voltage is kept constant,

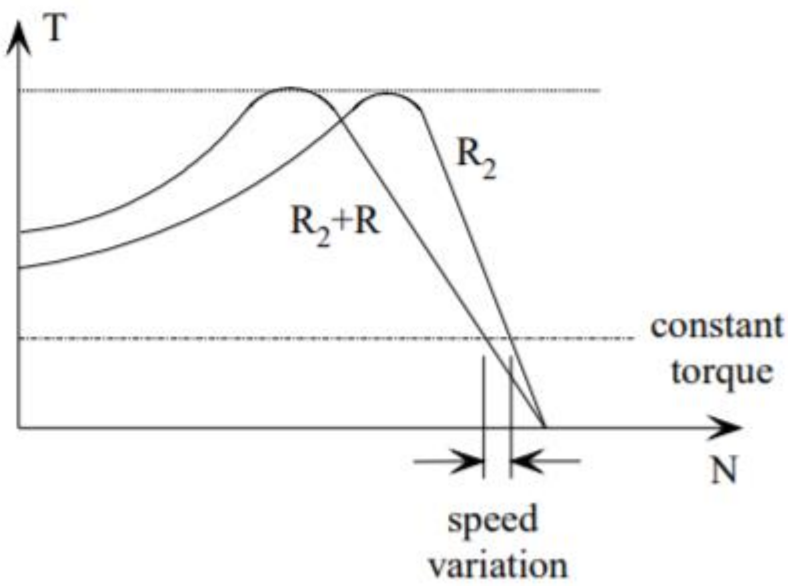
$$T \propto \frac{s}{R_2}$$

- For varying rotor resistance, an external resistance is introduced in rotor circuit through slip rings & brush arrangement. It is shown in figure
- The method is similar to armature voltage control of shunt motor

- *Advantages* - Simple, low cost, possible to get high starting torque
- *Drawbacks*

- With increase in rotor resistance, loss increases & efficiency decreases
- Here speed depends on R_2 & load





iv) <u>V/f control</u>

- Speed of Induction motor can be varied by varying supply frequency
- But from EMF equation of Induction motor, $\phi \alpha (1/f)$
- So when we decrease frequency to reduce speed, flux increases & causes saturation of core. The core losses also increases.
- If frequency is increased to increase speed, flux reduces & torque produced by motor reduces.
- So we have to maintain flux in the machine constant.

- From EMF equation, $\phi \propto (V/f)$
- By keeping V/f ratio constant, we can keep flux constant
- So instead of controlling frequency alone, *V* & *f* are varied simultaneously to keep *V/f* ratio constant
- Here motor is supplied from a variable frequency variable voltage supply
- T – speed characteristics of Induction motor with V/fcontrol is shown in figure
- *Advantages*
- Smooth speed control is possible
- Speed control above & below rated speed is possible
- *Drawbacks*
- A variable voltage variable frequency supply is required
- Costly & complicated